

idTech4 MD5 Tools

Blender 4.x / 5.x Import / Export / Retarget Add-on

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1. Overview

The idTech4 MD5 Tools add-on provides full import, export, and retargeting support for the Doom 3 / id Tech 4 .md5mesh, .md5anim, and .md5camera file formats inside Blender 4.x and 5.x. Both the original Version 10 format and the extended Version 12 format (with normals, tangents, and vertex colours) are supported, as are the Doom 3 BFG compiled binary caches .bmd5mesh and .bmd5anim.

Supported operations:

- Import .md5mesh / .bmd5mesh — skeletal mesh + armature
- Import .md5anim / .bmd5anim — animation onto an existing armature
- Export .md5mesh — skeletal mesh + armature
- Export .md5anim — one or more actions
- Export .md5mesh + Anim(s) — combined single-dialog workflow
- Import .md5camera — one or more camera files, each into its own named camera object with keyframed location, rotation, and FOV
- Export .md5camera — one or more camera actions to individual files with per-action cut filtering
- Auto-generate real, fully textured Blender materials on mesh import (needs the companion "idTech4 Materials" add-on)
- Animation Retargeter — re-apply animations after bone orientation edits
- Bone Collection Manager — control which bones are included on export

This add-on is one of six that make up the idTech4 Blender toolchain. It installs and runs entirely on its own, but shares its Base Directory / Mod Base / Materials Source settings with the others, and cooperates with them when they are present — see the idTech4 Blender Toolchain Overview document.

2. Installation

1. Open Blender and navigate to Edit > Preferences > Add-ons.
2. Click Install... and select idTech4_MD5_Tools.py.
3. Tick the checkbox next to "idTech4 MD5 tools - .md5mesh,md5anim, and md5camera Import/Export".
4. Close Preferences. The add-on is now active.

After installation the following entries appear:

- File > Import > idTech4 MD5 Mesh (.md5mesh, .bmd5mesh)
- File > Import > idTech4 MD5 Anim (.md5anim, .bmd5anim)
- File > Import > idTech4 MD5 Camera (.md5camera)
- File > Export > idTech4 MD5 Mesh (.md5mesh)
- File > Export > idTech4 MD5 Anim (.md5anim)
- File > Export > idTech4 MD5 Mesh+Anim(s) (.md5mesh,.md5anim)
- File > Export > idTech4 MD5 Camera (.md5camera)
- Properties > Data (armature) > Manage MD5 Export Bone Collection
- 3D Viewport > N-Panel > MD5 > MD5 Animation Retargeter
- 3D Viewport > N-Panel > idTech4 > Sources

NOTE: The Sources panel in the idTech4 sidebar tab is shared. Every idTech4 add-on ships an identical copy of it and only one copy is ever registered, so the panel looks and behaves the same no matter which subset of the add-ons you have installed.

3. Shared Settings

3.1 Source Directories

idTech 4 games do not ship their assets as loose files. Doom 3, Quake 4, Prey and The Dark Mod pack them into .pk4 archives; Doom 3 BFG packs them into .resources files. This add-on does not read inside either. It reads an ordinary directory tree on disk, and nothing else.

So before you set Base Directory, extract everything. A .pk4 is an ordinary zip archive — any zip tool will open one, and most will do it without renaming the file first. A .resources file is not a zip and needs a Doom 3 BFG extraction tool.

Extract every archive into one folder, preserving the paths stored inside them, so the result is the full idTech 4 directory structure:

```
<game>/
  base/           ← Base Directory points HERE
    def/          entityDef and model declarations
    dds/          precompressed textures
```

```

guis/
maps/           .map source files
materials/      .mtr material declarations
models/         .md5mesh, .md5anim, .ase, .lwo
skins/          .skin declarations
textures/       .tga / .png / .jpg source textures
generated/     BFG only: .bimage, .bmd5mesh,
                  .bmd5anim, .base, .blwo caches

```

IMPORTANT: Base Directory is the "base" folder ITSELF — the one that CONTAINS materials/, models/, textures/ and the rest — not the game folder above it, and not one of the folders inside it. Every path in a .mtr, .def, .skin or .map is written relative to this folder, so pointing one level too high or too low resolves nothing.

NOTE: A game ships several numbered archives (pak000.pk4, pak001.pk4, ... or game00.pk4, game01.pk4, ...) and the later ones patch the earlier ones. Extract them in ascending numeric order and let later files overwrite earlier ones — that is the same precedence the engine applies at run time. Extracting them in the wrong order silently leaves you with pre-patch assets.

For a mod or an expansion, extract its own archives into its own folder beside base — d3xp, darkmod, and so on — and set Mod Base Directory to that folder while leaving Base Directory on base. A mod supplies part of the tree and inherits the rest, and the add-ons search the two the same way the engine does.

NOTE: The extracted tree only has to contain what you actually intend to load. Sound, video and script archives can be left packed. The folders that matter are the ones in the diagram above.

Base Directory, Mod Base Directory and Materials Source live in a small JSON file under Blender's per-user config directory, not in any one add-on's preferences. They are edited from the 3D Viewport N-Panel, idTech4 tab, Sources panel, and every import dialog shows them read-only.

Setting	Required	Purpose
Base Directory	Yes	A standard Doom 3-style "base" folder — the one containing materials/, models/, textures/ and so on. Everything is resolved under it.
Mod Base Directory	No	An optional second tree searched BEFORE Base Directory, with Base Directory still searched behind it. This is the engine's own <code>fs_game</code> behaviour: a mod supplies part of the tree and inherits the rest.
Materials Source	No	One .mtr file, or a folder of them. Leave it blank and the "materials" subfolder under Mod Base (if set) and then Base Directory is used.

If Auto-Generate Materials is ticked and no Base Directory is configured, the import stops and offers a one-off setup dialog once the file and all the other options have been chosen:

Choice	Effect
Select Now	Type or browse a Base Directory (and optionally a Materials path) in the popup. "Set as default for future imports" is ticked by default and writes them to the shared config; untick it to use them for this import only.
Derive Automatically — just this time	Walk up from the imported file's own directory looking for a "materials" folder; Base Directory becomes that folder's parent. Nothing is saved.
Derive Automatically — and save settings	The same, but the derived paths are written to the shared config once they are actually known.
Skip — Don't Set Up Materials For This Import	Import with blank placeholder materials. Auto-Generate Materials is forced off for this one run.

NOTE: The popup's own Cancel button (and Escape) ends the import without importing anything. The gate only ever fires once per import, and only when it could actually accomplish something — it is skipped entirely when the paths already resolve, or when the "idTech4 Materials" add-on is not installed and enabled.

3.2 Scale

Every importer and exporter (mesh, anim, and camera) shares the same Transforms box. Scaling is OFF by default, so nothing is scaled unless you choose otherwise.

Option	Type	Default	Description
Scale Mode	<i>Enum</i>	None	None (no scaling), the idTech4 ↔ Blender unit preset, or Custom Factor. idTech4 units are inches and Blender's default unit is metres, so on import the preset multiplies by 0.0254 and on export by 39.3701.
Scale Factor	<i>Float</i>	1.0	An arbitrary uniform multiplier. Only shown when Scale Mode is Custom Factor. Range 0.0001 – 10000.

3.3 Rotation

Rotation is a pure rotation about the Z axis, applied to every position and orientation the operation reads or writes.

Choice	Meaning
0° : No rotation	Nothing is rotated. The default.
90° : X to Y	Rotate +90° about Z — X-forward to Y-forward, i.e. idTech4 → Blender.
-90° : Y to X	Rotate -90° about Z — Y-forward to X-forward, i.e. Blender → idTech4.
180° : Flip	Rotate 180° about Z.

NOTE: The import and export dropdowns list the same four choices in a different order, so the option most likely to be wanted sits nearest the top on each side. The values themselves are identical.

4. Importing

4.1 Import MD5 Mesh

Use File > Import > idTech4 MD5 Mesh (.md5mesh, .bmd5mesh) to import a skeletal model. The importer creates a new Collection named after the file and places both the Armature object and all Mesh objects inside it. Text .md5mesh and Doom 3 BFG binary .bmd5mesh files are both accepted, and the dialog adapts to which one you picked.

Option	Type	Default	Description
Scale Mode / Factor	<i>Enum / Float</i>	None / 1.0	See Section 3.2.
Rotation	<i>Enum</i>	0°	See Section 3.3. Rotates all imported geometry and bone positions around Z.
Set mesh shading to Smooth	<i>Bool</i>	On	Enable smooth shading on every imported mesh — equivalent to Object > Shade Smooth.
V12 Don't Import Normals	<i>Bool</i>	Off	V12 text files only. Ignores stored normals; Blender derives them from geometry. Hidden for .bmd5mesh, which carries its own.
V12 Bone Local to Blender Custom Normals	<i>Bool</i>	Off	V12 text files only. Transforms stored bone-local normals to world space and applies them as Blender custom split normals. Greyed out while "V12 Don't Import Normals" is ticked.
Merge Verts by Distance	<i>Bool</i>	Off	After import, runs Merge by Distance on every mesh in the imported collection.
Merge Distance	<i>Float</i>	0.0001	Maximum vertex distance to merge. Only active when Merge Verts by Distance is

Option	Type	Default	Description
			enabled.
<code>Auto-Generate Materials</code>	<i>Bool</i>	On	Build real, fully textured materials from the .mtr tree instead of blank placeholders. Greyed out when the "idTech4 Materials" add-on is not installed and enabled.
<code>Material Mode</code>	<i>Enum</i>	Maximum	Which of the Materials add-on's fidelity modes to build with: Maximum, Good, Basic or Simple. A single model is cheap enough to build at full fidelity, which is why this defaults higher than the .map importer's.
<code>Parameters</code>	<i>Enum</i>	Dynamic	What to do with time, parm0..11, global0..7, sound and table lookups: Dynamic (drivers, animates with the timeline), Baked (folded once, no drivers) or Skip.
<code>Derive from model</code>	<i>Bool</i>	Off	Ignore the three paths below and derive them for this import by walking up from the imported file's own directory looking for a "materials" folder.
<code>Base / Mod Base / Materials</code>	<i>Display</i>	—	Read-only mirrors of what this import will use. Hover to see the full path. Edit them in the idTech4 tab's Sources panel.
<code>Load anims after</code>	<i>Bool</i>	Off	When the mesh import finishes, open the MD5 Anim import dialog immediately, already targeting the armature this import just built.

NOTE: The importer always imports the armature. It is not possible to import the mesh alone.

NOTE: Merge Verts by Distance is OFF by default. An MD5 mesh is stored with its UV seams and hard edges already split into separate vertices, exactly as the engine renders it, so welding them is a change to the model rather than a cleanup of an import artifact.

When materials are generated the importer switches every 3D viewport in the current screen to Material Preview shading, and writes a full build report to a Text datablock. The operator's status line names that datablock; the Materials add-on's own Report section also shows it.

4.2 Import MD5 Anim

Use File > Import > idTech4 MD5 Anim (.md5anim, .bmd5anim) to load one or more animation files onto an existing armature. Animations are imported to the armature of the currently active collection by default, or select a target armature from the dropdown before import. Multiple files can be selected in the file browser — each file creates one Action, and the first one imported becomes the armature's active action.

Option	Type	Default	Description
Target Armature	<i>Enum</i>	active armature	The armature object to receive the animation. Initialised from the active object, falling back to the armature of the current collection.
Scale Mode / Factor	<i>Enum / Float</i>	None / 1.0	Should match the scale used when the mesh was imported.
Rotation	<i>Enum</i>	0°	Should match the rotation used when the mesh was imported.
Assign fake user	<i>Bool</i>	On	Give every created action a fake user so unassigned actions are still saved with the .blend file.
Prepend Action Names	<i>Bool</i>	Off	Adds the text in the Prepend Text box to the front of every imported action name.
Prepend Text	<i>String</i>	<collection>_	Text prepended to action names. Initialised to the active collection's name with a trailing underscore.

IMPORTANT: If any of the action names that would be created already exist in the scene, the importer lists them and asks you to confirm before overwriting. Click Import a second time to proceed. In background mode there is no popup, so the import refuses instead — re-run with `confirm_overwrite=True`.

4.3 Import MD5 Camera

Use File > Import > idTech4 MD5 Camera (.md5camera) to import one or more camera animation files. Multiple files can be selected in the file browser. Each file creates its own camera object named MD5_Cam_<stem> (e.g. MD5_Cam_intro for intro.md5camera) with keyframed location, rotation, and field of view. The first imported file is set as the active scene camera.

Option	Type	Default	Description
Scale Mode / Factor	<i>Enum / Float</i>	None / 1.0	Applied to camera positions. Offsets below are never scaled.
Rotation	<i>Enum</i>	0°	Rotates camera position and orientation around Z during import.
X / Y / Z Offset	<i>Int</i>	0	Blender-space offset applied to the camera position after import, rotation and scale. Values are in Blender world units and are not scaled.
Clear Timeline Markers	<i>Bool</i>	On	Removes all existing timeline markers before placing the imported cuts. Only applied on the first file when importing multiple files.

NOTE: Blender stores FOV data in a different data block from actions, which complicates loading multiple .md5camera files simultaneously. To simplify this, each file is imported as a separate camera object (orange camera in the hierarchy) along with a separate camera data object (green camera in the hierarchy). An action is stored on both — it is important not to change these independently; they should always point to the same action name. Cuts are imported as timeline markers named after the action. The scene frame range is extended to cover all imported files. X/Y/Z offsets are useful to move the origin of the camera when importing and exporting — helpful when a scene has different map coordinates in Blender than in game.

Expected scene orientation is +X Forward, +Y Left, +Z Up — the same convention the .md5camera file itself uses. Camera positions are carried through unchanged; only the orientation is converted, since a Blender camera points down its own -Z while an MD5 camera points along +X. Use the Rotation preset if your scene is laid out on a different axis.

5. Exporting

IMPORTANT: When exporting a mesh or animation, all transformations must be applied prior to export or the resulting files will be incorrect/broken. To apply transformations select the object, press Ctrl-A and then select All Transforms. If applying all transforms seems to break your model or animation in Blender, you will need to update the model/animation so all transforms can be applied prior to export. This is the number one cause of problems when exporting to game.

NOTE: All three mesh/anim exporters switch the active object to Object Mode for the duration of the export and switch back afterwards. Blender does not write Edit Mode's working copy back to the mesh and armature datablocks until the mode is exited, and the exporters read those datablocks — without the round trip, exporting from Edit Mode wrote the model as it was when Edit Mode was entered, with no error and no warning. If the mode switch is refused (for example while another modal operator holds the context) the export reports the problem rather than writing stale data.

5.1 Export MD5 Mesh

Use File > Export > idTech4 MD5 Mesh (.md5mesh) with the armature selected. Only bones that belong to the MD5_export_bone_collection bone collection are written to the file — see Section 6 for how to manage that collection. The menu entry is greyed out when the scene has no armature. The filename is pre-filled with the active object's collection name.

Option	Type	Default	Description
Scale Mode / Factor	<i>Enum / Float</i>	None / 1.0	The export preset converts Blender metres to idTech4 inches. Use the inverse of the import scale to restore original units.
Rotation	<i>Enum</i>	0°	Rotates all exported data around Z. Select

Option	Type	Default	Description
			the inverse of the rotation used on import to restore MD5 space.
Bone Influence Limit	<i>Int</i>	4	Maximum number of bone weights per vertex written to the file. Range 1 – 8.
MD5 Version 12	<i>Bool</i>	Off	Exports in the extended V12 format including normals, MikkTSpace tangents, and vertex colours.
Split Sharp Edges	<i>Bool</i>	Off	Split vertices at edges marked sharp, or with differing corner normals, so hard edges are represented correctly in the vertex stream. Non-destructive — the Blender scene is never modified. Works for both v10 and v12.

5.2 Export MD5 Anim

Use File > Export > idTech4 MD5 Anim (.md5anim) with the armature selected. Each checked action in the list is exported to a separate .md5anim file in the chosen directory. The menu entry is greyed out when the scene has no armature carrying actions.

Option	Type	Default	Description
Scale Mode / Factor	<i>Enum / Float</i>	None / 1.0	Should match the scale used on export of the corresponding mesh.
Rotation	<i>Enum</i>	0°	Should match the rotation used on export of the corresponding mesh.
Compress Animation	<i>Bool</i>	On	Only writes components that change beyond the delta threshold. Static components are stored in baseframe only.
Delta Threshold	<i>Float</i>	0.0001	Minimum change required for a component to be considered animated. Only active when Compress Animation is on.
Bounds Scale	<i>Float</i>	1.0	Scales each frame's bounding box around its centre by this factor. The box centre stays fixed; only the half-extents are multiplied. Use to correct bounds that appear too small or too large in-engine.
Action list	<i>UIList</i>	—	Scrollable checklist of all actions in the scene, with a running count of how many are ticked. Use the All / None / Invert buttons to control the selection. The active action is listed first and pre-selected.
Change Prepend	<i>Bool</i>	Off	When enabled, modifies exported filenames using the Add or Remove mode below.
Add / Remove	<i>Radio</i>	Add	Add prepends the text box value to each

Option	Type	Default	Description
			filename stem; Remove strips it from the front.
<code>Prepend Text</code>	<i>String</i>	<code><collection>_</code>	Text to add to or remove from exported filenames. Initialised to the active collection name.

5.3 Export MD5 Mesh+Anim(s)

Use File > Export > idTech4 MD5 Mesh+Anim(s) (.md5mesh,.md5anim) to export the mesh and any number of animations in a single dialog. The mesh is always exported first. If no actions are selected the mesh is still saved and the exporter reports the omission without error. Unlike the standalone anim dialog, every action starts out ticked here.

NOTE: The mesh export options and animation export options share the same Scale and Rotation fields, so both halves of the export are always consistent by construction. Every other option behaves exactly as it does in Sections 5.1 and 5.2, Bounds Scale included.

5.4 Export MD5 Camera

Use File > Export > idTech4 MD5 Camera (.md5camera) to export one or more camera actions. A scrollable checklist lists all actions in the scene. Each selected action is evaluated frame by frame and written to its own .md5camera file in the chosen directory, named after the action.

The export operator reads the frame range directly from the action's own keyframe range, so each file covers exactly the frames that were animated — regardless of the scene frame range.

Option	Type	Default	Description
<code>Scale Mode / Factor</code>	<i>Enum / Float</i>	None / 1.0	Applied to camera positions before the offsets below.
<code>Rotation</code>	<i>Enum</i>	0°	Rotates camera position and orientation around Z during export.
<code>Frame Rate</code>	<i>Int</i>	24	Frame rate written into the .md5camera header. Does not affect the number of frames exported. Range 1 – 240.
<code>X / Y / Z Offset</code>	<i>Int</i>	0	Blender-space offset applied to camera positions after scale and before the export rotation. Not affected by the Scale option.
<code>Action list</code>	<i>UList</i>	—	Scrollable checklist of all actions, with a running count of how many are ticked. Use All / None / Invert. The active camera action is pre-selected.

Option	Type	Default	Description
<code>Export All Cuts</code>	<i>Bool</i>	Off	When off, only timeline markers whose name matches the action being exported are written as cuts. When on, all timeline markers are written to every exported file.

IMPORTANT: Export All Cuts is intended for single-action exports only. When exporting multiple actions, each file will receive every cut in the timeline, which is unlikely to be correct.

6. Managing the Export Bone Collection

6.1 Why a Bone Collection?

Production armatures frequently contain bones that are not part of the deformation skeleton — IK targets, pole targets, helper bones, twist controllers, and other control rig elements. The MD5 format only supports a flat hierarchy of deformation bones; exporting control bones will either corrupt the file or cause the game engine to reject it.

The add-on solves this with a dedicated bone collection named `MD5_export_bone_collection`. Only bones that are members of this collection are written to `.md5mesh` and `.md5anim` files. All other bones are silently excluded, and a child whose immediate parent is excluded is re-parented to the nearest exported ancestor, or made a root.

6.2 The Bone Collection Manager Panel

Select the armature, open the Properties editor, and click the Data tab (the armature icon). Scroll down to the Manage MD5 Export Bone Collection section. The panel provides five buttons:

Button	Action
Add Selected (Create collection if needed)	Adds all selected bones (in Edit or Pose mode) to <code>MD5_export_bone_collection</code> without removing any existing members. Always enabled — this is the button that creates the collection in the first place.
Remove Selected	Removes all selected bones from the collection. The bones themselves are not deleted.
Replace with Selected	Clears the collection completely and then adds all currently selected bones. Use this to redefine the export set from scratch.
Clear All	Removes all bones from the collection. No bones will be exported until new ones are added.
Select All Bones in Collection	Selects every bone in the collection in the viewport. Useful for reviewing which bones are included.

NOTE: Remove Selected and Replace with Selected are greyed out unless bones are selected and the collection already exists; Clear All and Select All Bones in Collection are greyed out until the collection exists.

6.3 Typical Workflow — Control Rig

A common scenario is a character that has both a deformation skeleton and a control rig layered on top. The steps below show how to restrict the export to deformation bones only.

1. Switch to Edit Mode on the armature (Tab).
2. In the Outliner or Bone Properties, identify the deformation bones. These are the bones the mesh weights reference — typically named without the CTRL_, IK_, or TGT_ prefixes used by control bones.
3. Select all deformation bones. A quick method: deselect all (Alt+A), then Shift-click each deformation bone, or use the Outliner to filter by name.
4. In the Bone Collection Manager panel, click Replace with Selected. The collection now contains exactly the bones you selected.
5. Click Select All Bones in Collection to verify the result — only the deformation bones should highlight.
6. Export as normal. The resulting .md5mesh and .md5anim files will contain only the bones in the collection.

IMPORTANT: If the MD5_export_bone_collection is missing or empty at export time, the exporter raises an error. Always verify the collection contains at least one bone before exporting.

6.4 Armature Hierarchy in the Outliner

After importing an MD5 mesh, the Outliner will show a structure similar to the following. The collection created by the importer contains both the armature and all mesh objects.

```
Scene Collection
├── marine ← Collection created on import
│   ├── marine_MD5_Armature ← Armature object
│   │   └── (Bone data)
│   │       ├── origin
│   │       ├── pelvis
│   │       ├── spine
│   │       ├── neck
│   │       └── ... (all deformation bones)
│   ├── mesh_body ← Mesh object
│   └── mesh_head ← Mesh object
```

Inside the Armature Data Properties the bone collection will appear as:

```
Bone Collections
├── MD5_export_bone_collection ← Bones in this collection are exported
```

When you add a control rig the outliner may look like this, and you will want to ensure control bones are excluded:

```
Scene Collection
└─ marine
    └─ marine_MD5_Armature
        └─ (Bone data)
            ├── origin          ← deformation - include
            ├── pelvis          ← deformation - include
            ├── IK_foot_L       ← IK target - EXCLUDE
            ├── CTRL_spine      ← control - EXCLUDE
            └─ TGT_head         ← pole target - EXCLUDE
            ...
```

7. Baking Animations for Export

7.1 Why Baking Is Required

The MD5 animation format stores explicit bone transforms for every frame. It has no concept of constraints, drivers, IK chains, or NLA tracks — only the final world-space pose at each frame number matters. If your Blender animation uses any of the following, the raw Action keyframes will not represent the actual deformation and must be baked first:

- Inverse Kinematics (IK) constraints
- Copy Rotation / Copy Location / Copy Transform constraints
- Stretch To or Damped Track constraints
- Driver-controlled bones
- NLA tracks blending multiple actions
- Physics or cloth simulations influencing the rig

Baking evaluates Blender's full constraint stack at every frame and writes the result as plain keyframes on a new Action. The exporter then reads those keyframes directly.

NOTE: If none of the above are used in an animation, baking is not needed and the animation can be exported with no additional steps.

7.2 Baking a Single Action

1. Set the scene frame range (frame_start and frame_end) to cover the animation you want to bake.
2. Select the armature object.
3. Switch to Pose Mode (Ctrl+Tab or the mode menu in the 3D Viewport header).
4. Select all bones: A.

5. Open the Object menu in the 3D Viewport header (or press Space and search) and choose Animation > Bake Action...
6. In the Bake Action dialog, configure the options described below, then click OK.

Bake Action dialog settings for MD5 export:

Setting	Recommended	Notes
Frame Start / End	Match scene	Should span the full range of the animation.
Step	1	Step of 1 produces a keyframe for every frame. Larger values reduce resolution.
Only Selected Bones	On	Bake only the bones in your selection. If you want to bake the full skeleton, select all bones first.
Visual Keying	On	CRITICAL: captures the final constraint-resolved pose. Without this, constraints are ignored.
Clear Constraints	Off	Leave constraints in place so the rig remains editable. The baked action is independent.
Clear Parents	Off	Keep the bone parent hierarchy intact.
Overwrite Current Action	Off	Off creates a new Action. On replaces the current one. Off is safer.
Bake Data	Pose	Must be set to Pose for armature / animation baking.

IMPORTANT: Visual Keying must be enabled. Without it, IK, constraints, and drivers are not evaluated and the baked action will contain only raw keyframe values, which will be wrong for any constrained bone.

7.3 Baking Multiple Actions (NLA Workflow)

When a character has several animations managed through the NLA Editor, each action must be baked individually. The recommended approach is:

1. Open the NLA Editor (Shift+F4 or the editor type menu).
2. Locate the action strip you want to bake. Click the Push Down button (the down-arrow icon next to the action name in the Action Editor, or right-click > Push Down in the NLA Editor) to push the action into the NLA stack as a strip.
3. Solo the strip you want to bake by clicking the star icon on the strip. Only that strip will be evaluated during baking.
4. Return to the 3D Viewport, select all bones in Pose Mode (A), and run Animation > Bake Action... with Visual Keying on.
5. Rename the new baked action (in the Action Editor header) to something descriptive like marine_run_baked.
6. Disable the solo star on that strip, then repeat from step 3 for the next action.

NOTE: After baking, the new actions appear in the action list of the MD5 Anim export dialog automatically. You can use the All / None / Invert buttons or tick individual actions to control exactly which ones are exported.

7.4 Verifying a Baked Action

Before exporting, verify the bake was successful:

1. In the Action Editor (Shift+F5 or Dope Sheet > Action Editor mode), select the baked action from the drop-down.
2. Scrub the timeline. The mesh should deform correctly on every frame without any constraints being active.
3. If the pose looks wrong, check that Visual Keying was on, the correct frame range was used, and that the correct bones were selected during baking.

7.5 Exporting After Baking

Once your baked actions are ready:

1. Select the armature.
2. Open File > Export > idTech4 MD5 Anim (.md5anim).
3. In the action list, tick only the baked actions you want to export. Untick any source (pre-bake) actions to avoid exporting them.
4. If your baked actions have a common prefix (e.g. marine_), use Change Prepend with Remove to strip it from the output filenames.
5. Click Export MD5 Anim.

8. Animation Retargeter

8.1 Purpose

When you adjust bone orientations on an armature that already has imported animations — for example, to fix an axis alignment issue after import — all existing animations will appear broken because the stored pose data no longer matches the new bone orientations.

The Animation Retargeter corrects this. It takes a snapshot of the original bone orientations, lets you make your edits, and then recomputes every keyframe in every affected action so that the visual pose is preserved with the new bone orientations. Note that this is a simple retargeter designed to correct simple rotation edits of an armature, not full retargeting from one armature or model to another. It was primarily designed for editing original idTech4 assets — when importing many original models the bones have orientations that can be annoying to work with.

8.2 Location

The retargeter panel is found in the 3D Viewport N-Panel (press N to open it) under the MD5 tab — its own tab, kept separate from the shared idTech4 tab that carries Sources. Select the armature before using any retargeter operator; the panel is hidden entirely when the active object is not an armature.

8.3 Retargeting Workflow

1. Import an MD5 mesh and all animations using this mesh. Any animation not loaded/retargeted will no longer work after bone orientation changes are made.
2. Take Snapshot — First select the armature in the 3D Viewport, then click Snapshot Rest Pose. This records the current rest-pose bone matrices. The snapshot is stored on the armature object and persists across saves. The panel then reports how many bones it holds.
3. Edit Bone Orientations — You can auto align bones and/or manually edit bone rotations and lengths. The two auto-align buttons only enable while you are in Edit Mode.
4. Clicking Auto Realign Bones will automatically point a parent bone to the child bone with the most descendants. If all child bones have the same number of descendants, the parent will point to the median location of all child bones. Leaf bones (bones with parents but no children) are pointed in the same direction as their parent. If the Adjust Bone Lengths checkbox is enabled, the tail of a parent bone will be adjusted to touch the head of the child it points to. The Only Realign Selected checkbox restricts this to selected bones only.
5. Clicking Adjust Lengths Only will not change any bone rotations, but will change the bone length so the tail of the bone touches the head of the child bone with the most descendants. (MD5 files do not store bone lengths. If the armature rotations are already correct, this is an easy way to visualise bone parenting.) This button also obeys the Only Realign Selected checkbox.
6. Manually edit bone orientations if needed. Switch to Edit Mode and adjust bone roll angles or orientations as needed. Do not change bone locations, or add/remove bones at this stage. (Note that clicking Auto Realign Bones after manually editing a bone rotation may overwrite any manual changes.)
7. Retarget — Return to Object Mode and click Retarget Animation(s). Each action receives a backup action named <action>_pre_retarg, and then every keyframe is recomputed using the correction matrix derived from the old and new bone orientations.
8. Verify — Scrub the timeline to verify poses look correct. If they do not, Restore Backup puts the previous action data back.
9. Cleanup — Once satisfied, click Delete Backups to remove the stored pre-retarget actions, and Clear Snapshot to drop the rest pose snapshot.

8.4 Retargeter Options

Option	Type	Default	Description
Adjust Bone Lengths	<i>Bool</i>	On	When enabled, bone lengths are changed so a parent's tail touches its child's head when Auto Realign Bones is pressed. Obeys Only Realign Selected.
Only Realign Selected	<i>Bool</i>	Off	Restricts Auto Realign Bones and Adjust Lengths Only to the currently selected bones. Unselected bones retain their original orientation.
Retarget All Actions	<i>Bool</i>	On	When on, all actions in the scene are processed. Turn it off to retarget only the currently active action.

The Utilities box at the bottom of the panel holds three buttons, each enabled only when it has something to do:

Button	Action
Restore Backup	Put the pre-retarget keyframes back on the active action.
Delete Backups	Delete every *_pre_retargt action in the file.
Clear Snapshot	Discard the stored rest-pose snapshot.

IMPORTANT: Always take a snapshot before editing bone orientations. If you edit bones without taking a snapshot first, there is no reference to compare against and retargeting will produce incorrect results. The Retarget button stays disabled until a snapshot exists.

9. Tips and Troubleshooting

'Corrupted' Exports

If a mesh or animation appears OK in Blender but is 'corrupt' in game, or when exported as .md5 and re-imported to Blender, there is an excellent chance that not all transformations were applied before exporting the mesh/animation. To apply transformations, with the object selected, press Ctrl-A and then select All Transforms. If this seems to break your model or animation then it will need to be updated so all transforms can apply correctly before export.

Exporting While Still in Edit Mode

This is handled for you — the exporters round-trip through Object Mode so Blender flushes the edit cage back to the datablocks first. If the round trip cannot be made, the export is refused with a message rather than quietly writing the pre-edit model.

Scale and Rotation Consistency

When importing an animation, use the exact same Scale and Rotation values used when importing the mesh. When exporting a Mesh, Animation, or Camera the Scale and Rotation settings are not necessarily the same as the import settings: they should be the inverse. If a mesh was imported with the idTech4 → Blender scale preset, use the Blender → idTech4 preset to export it at its original size; if it was imported with a rotation of 90° (X to Y), use -90° (Y to X) on export. When exporting an animation, the scale and rotation values should match the values used when exporting the mesh.

Action Names and File Names

By default each action is exported to a file named after the action. Use the Change Prepend options in the export dialog to add or strip a common prefix. For example, if all your baked actions are named marine_idle, marine_run, marine_pain, you can strip the marine_ prefix so

the output files are `idle.md5anim`, `run.md5anim`, `pain.md5anim` — matching the filenames the game expects.

Empty Bone Collection

If you receive an error on mesh or anim export stating that no bones were found, the `MD5_export_bone_collection` is missing or empty. Open the Bone Collection Manager panel on the armature and use Add Selected or Replace with Selected to populate it.

Normals and V12

Version 12 exports include per-vertex normals and tangents computed with the MikkTSpace algorithm (the same standard used by most modern game engines). If round-tripped normals look wrong, try enabling Split Sharp Edges in the export dialog so hard edges are represented correctly in the vertex stream.

Nothing Resolves At All

If every lookup fails — no materials found, no models found, no `.def` files found — the usual cause is that the game's assets are still inside their `.pk4` or `.resources` archives. These add-ons cannot read into an archive; everything has to be extracted to a real directory tree first. The second most common cause is Base Directory pointing at the game folder rather than at the base folder inside it: it must be the folder that directly contains materials/, models/ and textures/.

Materials Look Flat White

A model whose materials all render flat white usually means the material decls could not be found. Check that Base Directory (or Mod Base) actually points at a tree containing the right `.mtr` files, and read the build report the import writes to the Text Editor — it names every material that fell back to a placeholder and why. Material names are matched the way the engine matches them: backslashes converted to slashes, lowercased, and truncated at the last dot, so a surface spelled with either separator resolves to the same decl.

Binary `.bmd5mesh` / `.bmd5anim`

Doom 3 BFG ships compiled binary caches of its models under `generated/`. These are accepted by both importers and can be mixed freely with text files in one multi-file selection. They carry their own normals, so the two V12 normal options are hidden when a `.bmd5mesh` is selected. There is no binary exporter — export always writes text.

MD5 Camera Overview

Each `.md5camera` file stores a per-frame camera path with position, orientation (as a packed unit quaternion), and field of view in degrees. Cut frames mark scene transitions. On import, each file becomes a separate Blender camera object named `MD5_Cam_<stem>` so multiple camera animations can coexist in the same scene without overwriting each other.

Camera Coordinate System

Both the file and the expected Blender scene use +X Forward, +Y Left, +Z Up, and camera positions cross between them unchanged. Only orientation is converted, because a Blender camera looks down its own -Z axis while an MD5 camera looks along +X. If your scene is laid

out on a different axis, use the Rotation preset — for cameras it is applied as a Z rotation to both position and orientation. X, Y and Z offsets are in Blender world units and are never scaled.

Working with Multiple Camera Files

When importing multiple .md5camera files at once, select all of them in the file browser before clicking Import. Each file creates its own MD5_Cam_<stem> object. The Clear Timeline Markers option only fires on the first file so that cuts from subsequent files are added alongside existing ones rather than wiping them.

When exporting, use the action checklist to select exactly which camera actions to export. Each selected action is written to a separate file named after the action. Only timeline markers whose name matches the action are written as cuts, so each file receives only its own cuts.

Camera Import Errors

The .md5camera parser handles Windows CRLF line endings and // comments automatically. If a file fails to parse, open it in a text editor and verify that numFrames, frameRate, and numCuts are present and spelled correctly, and that the cuts { } and camera { } blocks are properly closed.

Merge by Distance After Import

Some MD5 meshes have duplicate vertices at UV seams. Enabling Merge Verts by Distance with a small threshold (0.0001) on import will weld these without affecting the visible geometry — but remember this changes the model away from what the engine renders, which is why it is off by default.